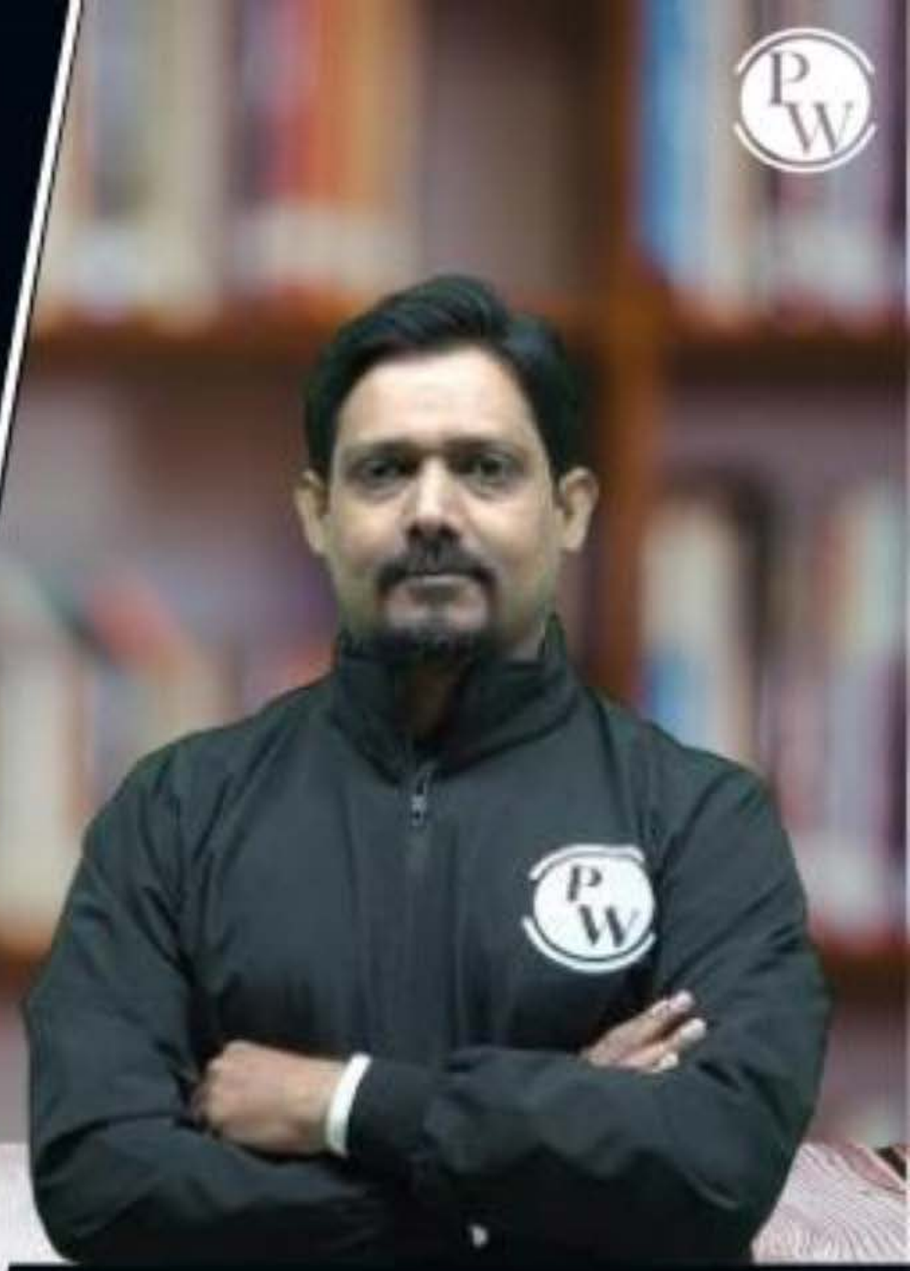
Computer Science & IT

Data Structure & Programming



Linked List

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



- Topic — Node* getnode(int x)
- Topic — Node* Build123()
- Topic — Display()
- Topic — Count()
- Topic — Adding 5 to each Number

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

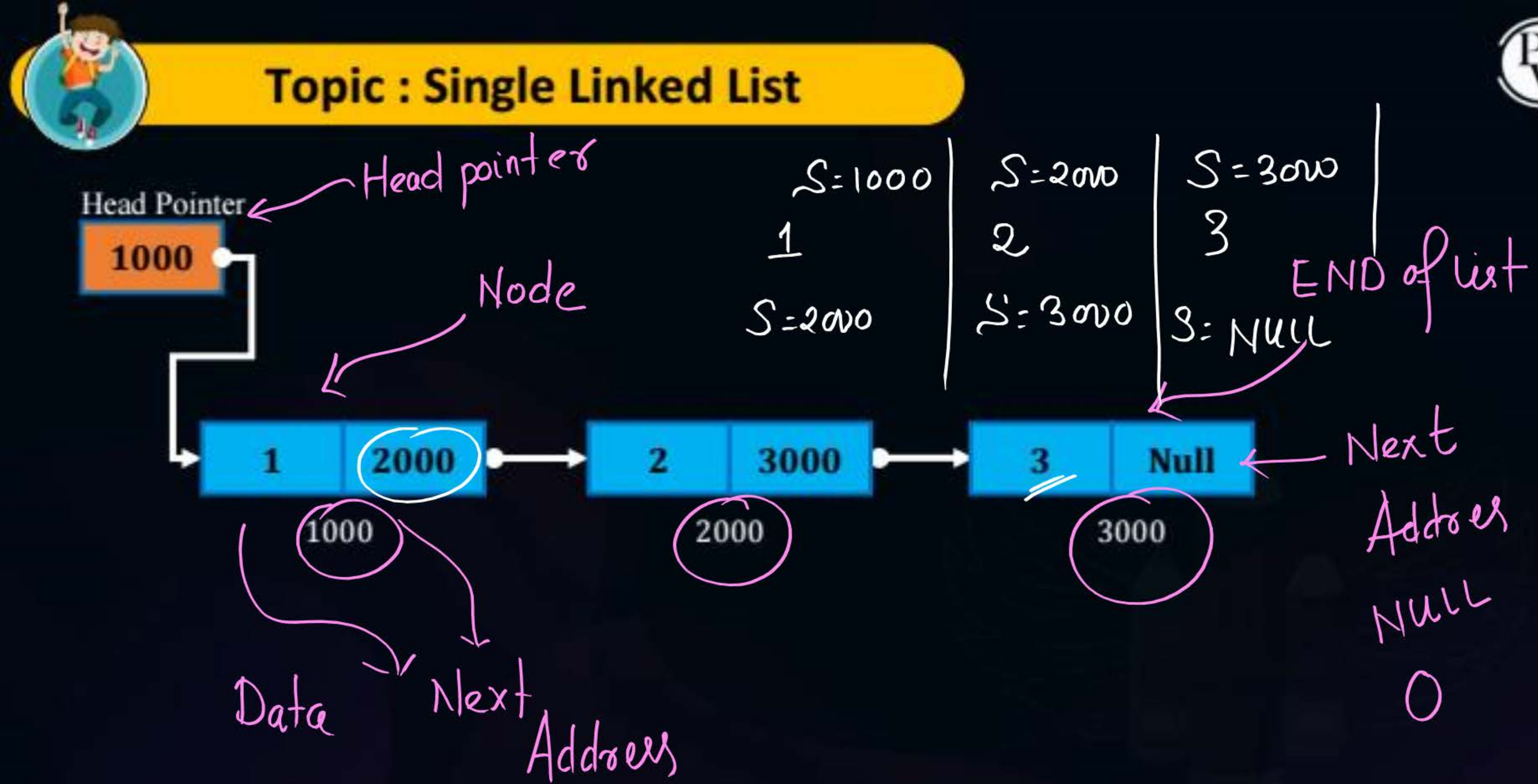
print begin, print Last Node

Add Begin Add end

practice problem

Slide

Topic : Single Linked List





Topic : Single Linked List

```
void display ( Node * s) {
    while ( s != NULL) {
        printf ("%d", s->data);
        s = s->next;
    }
}
```

Slide



Topic : Question

Consider the following function

```
int dosmothing ( struct node * q ){
int c = q -> data ;
if (q==null) return 0;
if (q -> link == NULL) return c;
else
c = c + dosmothing (q = q -> link) ;
return c ;
}
```

The above function run on the linked list that contain 1, 2, 3, 4, 5, 6, 7, 8. What will the content of the linked lists & the return value?

(A) 1, 2, 3, 4, 5, 6, 7, 8 return value 28

(B) 2, 1, 4, 3, 6, 5, 8, 7 return value 36

(C) 1, 2, 3, 4, 5, 6, 7, 8 return value 36

(D) 2, 1, 4, 3, 6, 5, 8, 7 return value 28

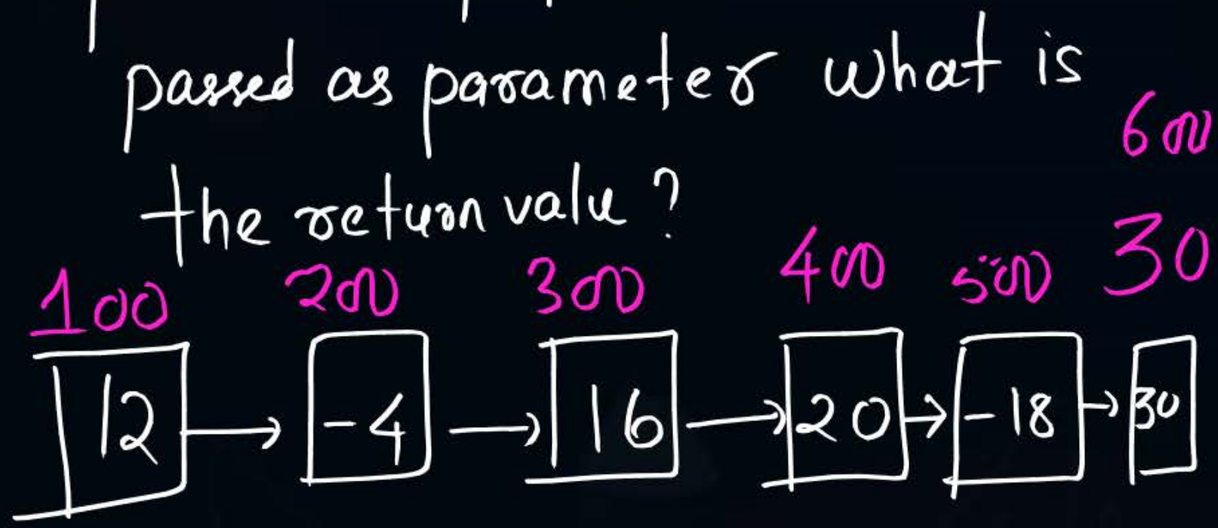


Topic : Single Linked List

```
int fun(Node *s){
    int x=0;
    while(s!=NULL){
        x= x+s->data;
        s= s->next;
    }
    return x;
}
```

12
− 4
16

if the last first Node is passed as parameter what is the return value?



74
18
56

Slide

Topic : Question

typedef struct node{
int data,
struct node *link;
}Node.



What would be the output after the following sequence of steps?

```
struct node *p;
p = first → link → link → link;
p → link → link = p;
first → link → link = p → link;
printf ("%c", first → link → link →link → data);
```

(a) a (b) b (c) c (d) d

p = 400

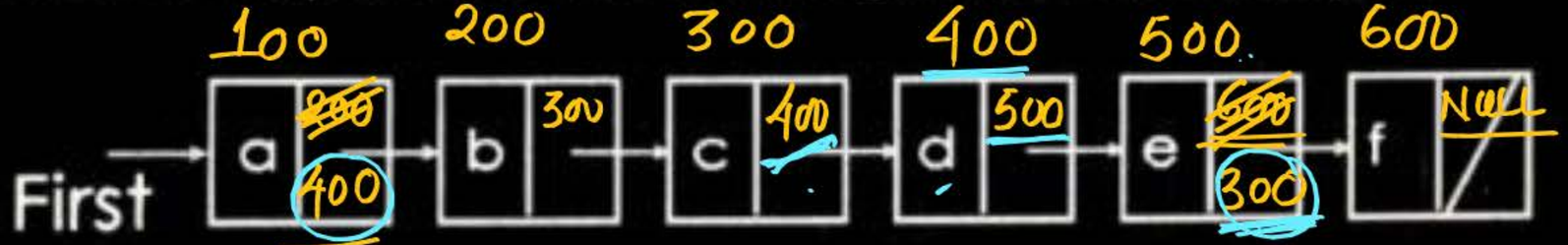
200 500 400

[d]

Slide

Topic : Question

02. What is the output after the following sequence of steps?




```
struct node *P;
(i)  P = first → link → link;
(ii) P → link → link → link = P;
(iii) first →link = P → link;
(iv) printf ("%c", first → link → link → link → link → data);
```

P = 300

400 500 300 400

d

(a) a (b) b (c) c (d) d

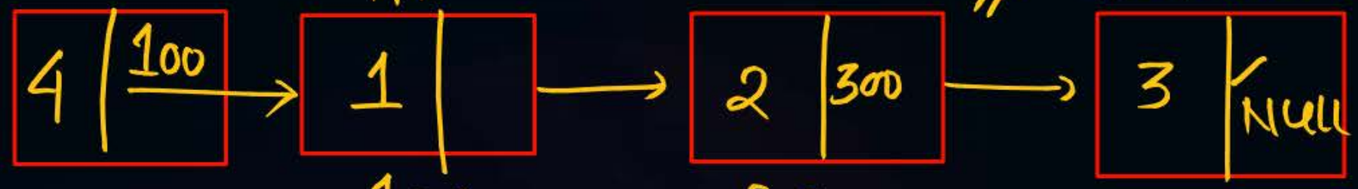
Slide

Topic : Single Linked List

Head pointer will change

Node * Addbegin (Node * S, int x) {

Insert at begining of Nonempty List



p→next will this change

Head pointer?

```
# include<stdlib.h>
# include<stdio.h>

int main(){
  Node * HP;
  HP= Build123( );
  HP= Addbegin( HP, 4);
}
```

```
Node * p = getnode(x);
p→next = S; ✓
return p;
}
```

if list is NULL Addbegin constant will work or Not?

Slide



Topic : Single Linked List

Adding a Node at end of list



(Non Empty)

```
void Addend(Node *S, int x){
    Node* p=getnode(x);
    while(S→next!=NULL)
        S=S→next;
    S→next = p;
}
```

Addend (HP. 6)

Linear time

↑ Size of list dependent

Slide



Topic : Single Linked List

Global version HP



without Node*

HP make it global variable

```
Node* HP;
int main(){
    HP = Build123()
```

No need to write Node* as return value

```
void Addbegin(int x){
    Node* p = getnode(x);
    p->next = HP;
    HP = p;
}
```

← global variable

it will update.

Slide



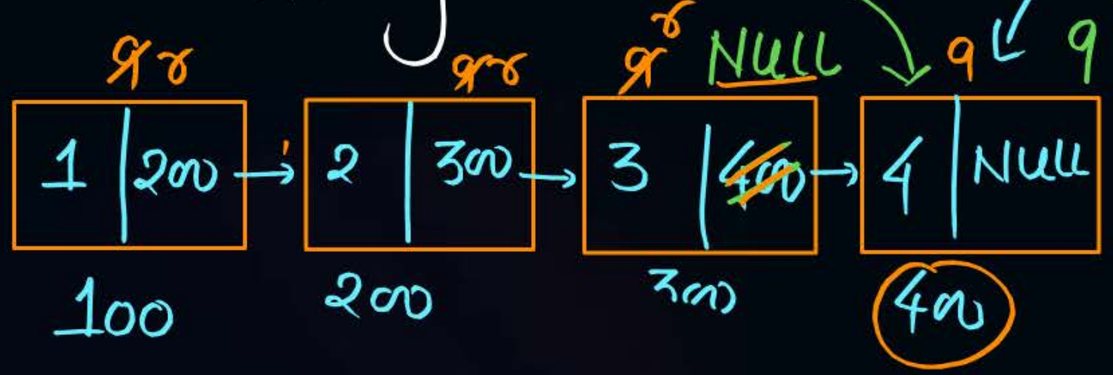
Topic : Single Linked List

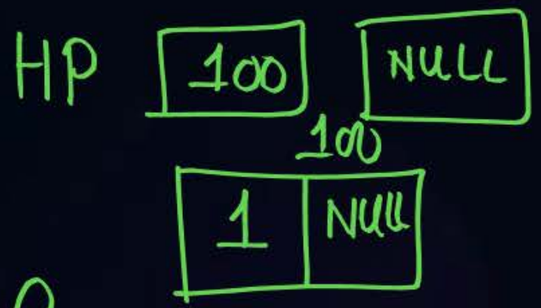
HW - global HP - Every case write Addend.

Slide

Topic : Single Linked List

Delete the Last Node from HP global





free(HP) free(100) HP=NULL

```
free(q);
r->next=NULL;
}
```

```
void deleteend(){
  Node *q=HP, *r;
  if(HP==NULL) //Empty list
     return;
  else if(HP->next==NULL){
                  // one element
     free(HP);
     HP=NULL;
  }
  else { while(q->next!=NULL)
       { r=q;
         q=q->next;
       }
```

Slide



Topic : Single Linked List

- traverse till second Last Node
- Delete Last Node
- Second Last Node Next will be assigned to NULL

Slide



Topic : Single Linked List

H.W

What is the output of following function if Linked list contain 1 → 2 → 3 and Head pointer is passed to foo function.

```
void foo ( Node * s) {
    if (s) {
        foo (s→next);
        printf("%d", s→data);
        foo (s→next);
    }
}
```

Slide



2 mins Summary

- Topic
- Topic
- Topic
- Topic
- Topic

Slide

THANK - YOU